

Lesson 6 Review: Point Estimation

BSTA 551: Statistical Inference

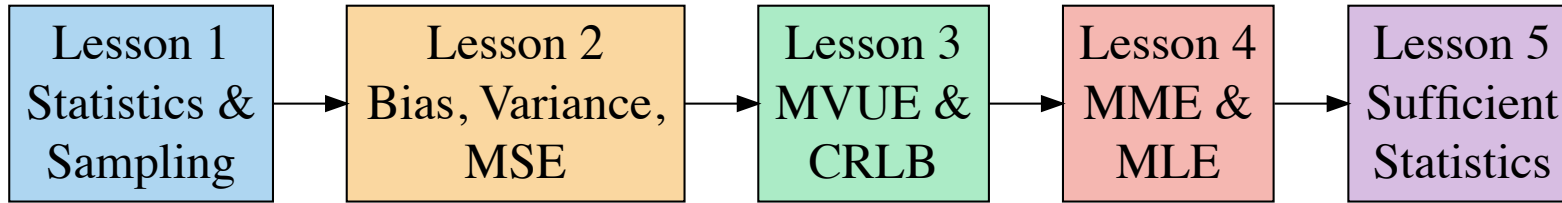
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2026-01-21

Review: The Big Picture of Point Estimation

(Lessons 1-5)

Where We've Been: The Journey So Far



Today: Tie it all together and review key concepts.

The Central Problem of Statistical Inference

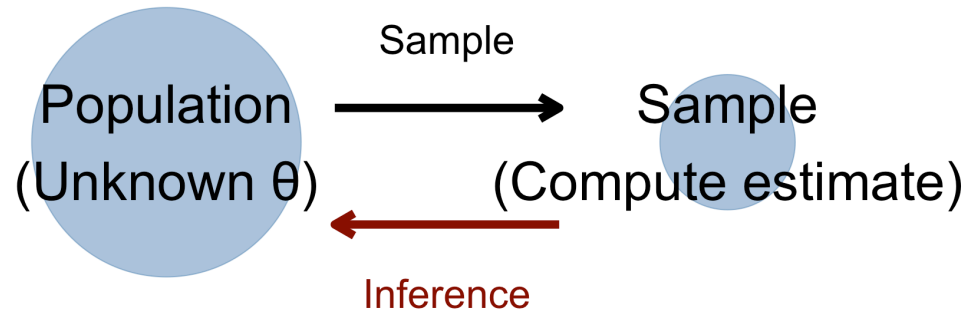
We observe data $X_1, X_2, \dots, X_n$ from some distribution with unknown parameter θ . Usually they are identically and independently distributed (i.i.d.), so they make a “random sample.”

The Question: How do we learn about θ from the data?

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The Two Key Questions in Point Estimation

Question 1: How do we **construct** estimators?

- Method of Moments (MME)
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- Bias (accuracy)
- Variance (precision)
- Mean Squared Error (overall quality)
- Efficiency (comparison to optimal)

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Question 3: How do we find the **best** estimator?

- MVUE: Minimum Variance Unbiased Estimator
- Sufficient statistics: Data reduction without information loss

Part 1: Properties of Estimators

The Fundamental Tradeoff: Bias vs. Variance

! Key Definitions

- **Bias:** $\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$ (systematic error)
- **Variance:** $\text{Var}(\hat{\theta}) = E[(\hat{\theta} - E(\hat{\theta}))^2]$ (random error)
- **MSE:** $\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + \text{Bias}^2$

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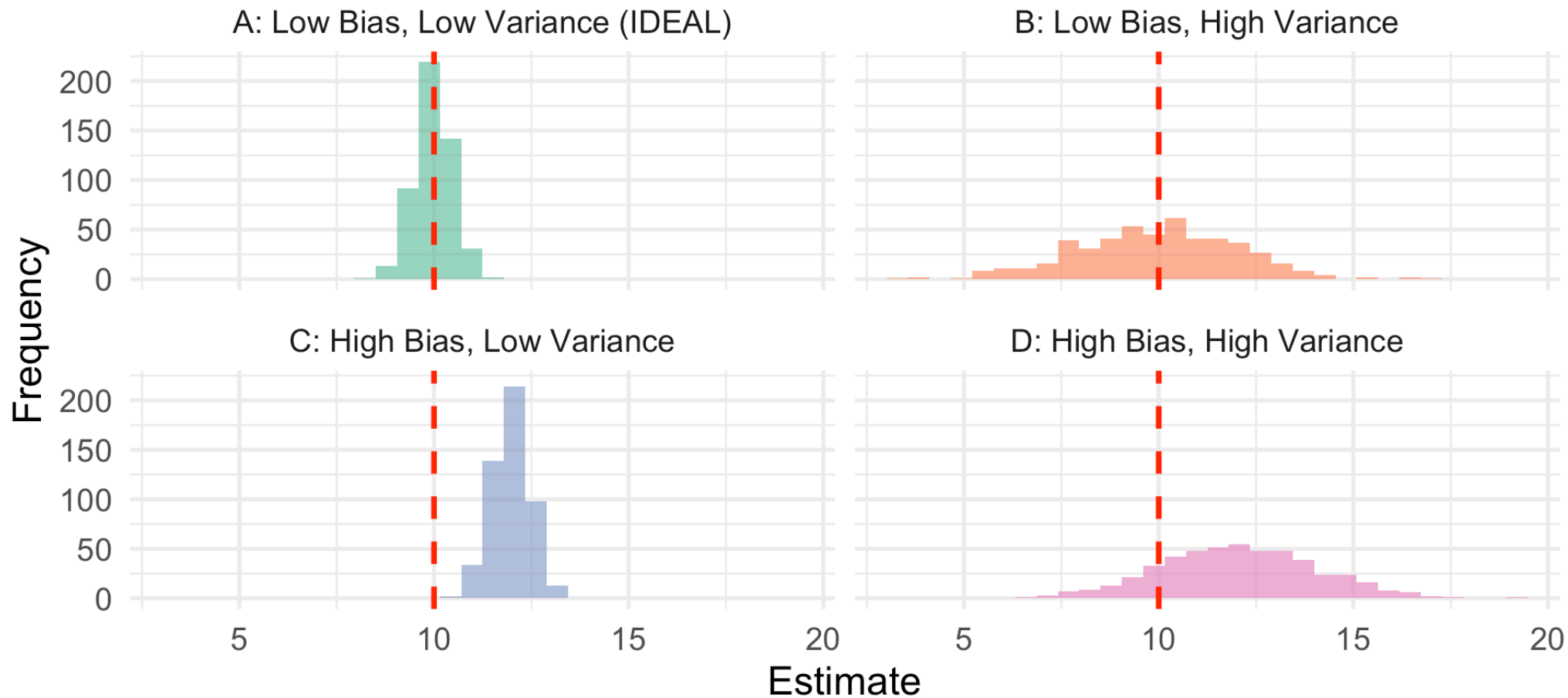
The key insight: $\text{MSE} = \text{Variance} + \text{Bias}^2$

This means a **biased** estimator can be better than an unbiased one if its variance is low enough!

Visual Intuition: Bias and Variance

The Four Scenarios of Bias-Variance

Red dashed line = true parameter $\theta = 10$



Medical Example: Estimating Treatment Effect

Scenario: Estimating the true cure rate p of a new cancer treatment.

Two estimators:

- **Standard:** $\hat{p}_1 = X/n$ (sample proportion, unbiased)
- **Add-Two:** $\hat{p}_2 = (X + 2)/(n + 4)$ (biased toward 0.5)

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Understanding the “Add-Two” Estimator: The Intuition

What does it do? It acts like you observed 2 extra successes and 2 extra failures before seeing the data.

$$\hat{p}_2 = \frac{X + 2}{n + 4} = \frac{\text{successes} + 2}{\text{trials} + 4}$$

Why +2 in numerator and +4 in denominator?

- Adding 2 successes and 2 failures = adding 4 total “pseudo-observations”
- This “pulls” the estimate toward $2/4 = 0.5$ (the middle)
- The pull is stronger when n is small, weaker when n is large



Why Does the Add-Two Estimator Help?

The Problem with Standard Estimator:

- If you observe 0 successes in 10 trials, $\hat{p}_1 = 0/10 = 0$
- But is the true p really exactly 0? Probably not!
- Small samples give extreme estimates that are likely wrong

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How Add-Two Helps:

- With 0 successes in 10 trials: $\hat{p}_2 = (0 + 2)/(10 + 4) = 2/14 \approx 0.14$
- This is more “conservative” – it hedges against extreme estimates
- **Shrinkage toward 0.5** reduces variance at the cost of small bias

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Bayesian Connection

The add-two estimator is the **posterior mean** under a $\text{Uniform}(0,1)$ prior on p . It incorporates prior belief that extreme values of p (near 0 or 1) are less likely.



Comparing the Two Estimators: Simulation

For $p = 0.2$ (a fairly extreme value), let's compare the two estimators:

► Code

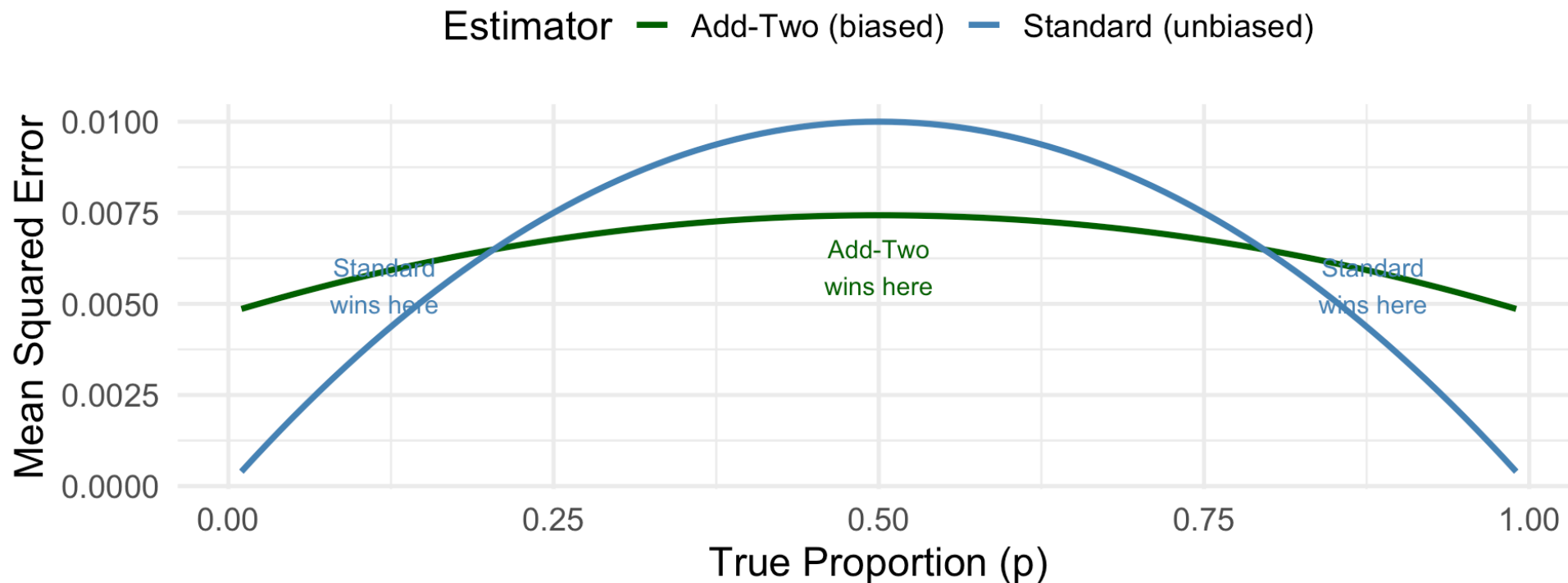
```
# A tibble: 4 × 3
  Metric      Standard `Add-Two`
  <chr>      <dbl>      <dbl>
1 Mean      0.199      0.241
2 Bias     -0.00101     0.0405
3 Variance  0.0065      0.00483
4 MSE       0.0065      0.00647
```

At $p = 0.2$, the MSEs are similar — we're in the “crossover” region where neither dominates!

When Does Add-Two Win? MSE Comparison

MSE Comparison: When Does Bias Help?

Add-Two wins near $p = 0.5$ (small bias);
Standard wins for extreme p (near 0 or 1)



Key insight: The add-two estimator wins **near** $p = 0.5$ where its bias is small, but loses at **extreme** p values where shrinkage toward 0.5 hurts!

What Properties Do We Want in an Estimator?

Property	Definition	Why It Matters
Unbiased	$E(\hat{\theta}) = \theta$	No systematic error
Low Variance	$\text{Var}(\hat{\theta})$ small	Precise estimates
Consistent	$\hat{\theta} \xrightarrow{P} \theta$	Works with enough data
Efficient	Achieves CRLB	Best among unbiased

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The Hierarchy of Goals

1. **Consistency** is the bare minimum
2. **Unbiasedness** is nice but not essential as long as bias shrinks as $n \rightarrow \infty$
3. **Efficiency** (low variance) is what really matters for precision



Part 2: Constructing Estimators

Two Systematic Methods: MME and MLE

Method of Moments (MME):

- Match population moments to sample moments
- Simple algebra, often intuitive
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Maximum Likelihood (MLE):

- Find θ that maximizes $P(\text{data}|\theta)$
- More complex, but has optimal properties
- Asymptotically efficient, unbiased, Normally distributed!

The MME Recipe

! Method of Moments: The Algorithm

1. Express the first k population moments as functions of θ : $E(X), E(X^2), \dots$
2. Set each equal to its sample counterpart: $\bar{X}, \frac{1}{n} \sum X_i^2, \dots$
3. Solve for θ

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Example: Exponential(λ)

- Population moment: $E(X) = 1/\lambda$
- Set equal: $1/\lambda = \bar{X}$
- Solve: $\hat{\lambda}_{MME} = 1/\bar{X}$

The MLE Recipe

! Maximum Likelihood: The Algorithm

1. Write the likelihood: $L(\theta) = \prod_{i=1}^n f(x_i; \theta)$
2. Take the log: $\ell(\theta) = \sum_{i=1}^n \ln f(x_i; \theta)$
3. Differentiate and set to zero: $\ell'(\theta) = 0$
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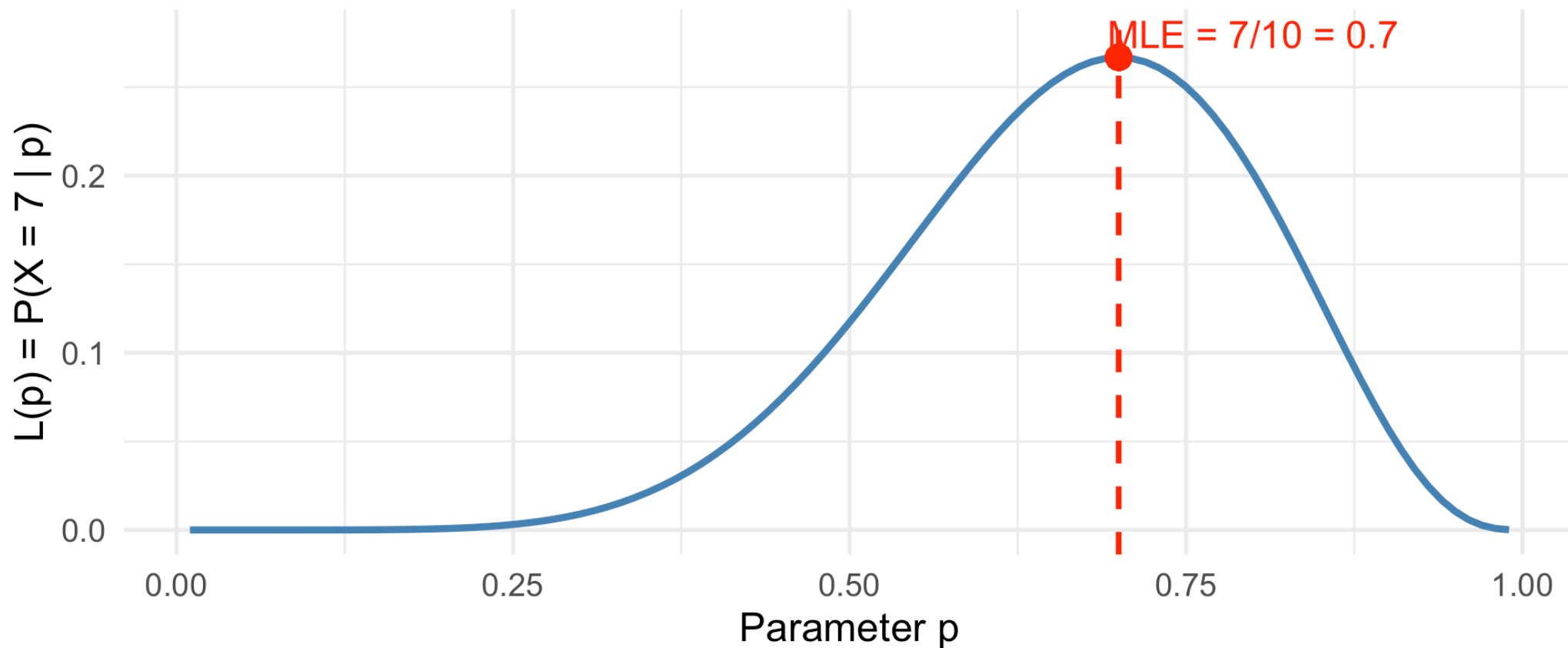
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Intuition: The MLE asks “What parameter value makes our observed data most probable?”

Visual Intuition: What Does the MLE Do?

Likelihood Function: 7 Successes in 10 Trials

MLE finds the peak at $p = 0.7$



The MLE finds where the likelihood is **highest** — the most “likely” parameter value.

When Do MME and MLE Agree?

Distribution	MLE	MME	Same?
Bernoulli(p)	$\bar{X}$	$\bar{X}$	✓
Normal(μ , known σ^2)	$\bar{X}$	$\bar{X}$	✓
Exponential(λ)	$1/\bar{X}$	$1/\bar{X}$	✓
Poisson(μ)	$\bar{X}$	$\bar{X}$	✓
Uniform[0, θ]	$\max(X_i)$	$2\bar{X}$	✗

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⚠ Key Insight

When they differ, the MLE is usually preferred (especially for large n) because it's asymptotically efficient.

MLE Properties: Why It's Often the Best Choice

! Three Key Properties of MLEs (Devore 7.2)

1. **Invariance:** If $\hat{\theta}$ is the MLE of θ , then $g(\hat{\theta})$ is the MLE of $g(\theta)$
2. **Consistency:** $\hat{\theta}_{MLE} \xrightarrow{P} \theta$ as $n \rightarrow \infty$
3. **Asymptotic Efficiency:** For large n , $\hat{\theta}_{MLE} \sim N\left(\theta, \frac{1}{nI(\theta)}\right)$

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In plain English:

- You can transform MLEs freely (invariance)
- MLEs “work” given enough data (consistency)
- MLEs are approximately the **best** for large samples (efficiency)

Consistency: What Does It Really Mean?

! Definition: Consistency (Devore 7.1, Chihara 6.3.4)

An estimator $\hat{\theta}_n$ is **consistent** if:

$$P(|\hat{\theta}_n - \theta| < \epsilon) \rightarrow 1 \quad \text{as } n \rightarrow \infty$$

for any $\epsilon > 0$, no matter how small.

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Intuition: With enough data, your estimate will be arbitrarily close to the truth.

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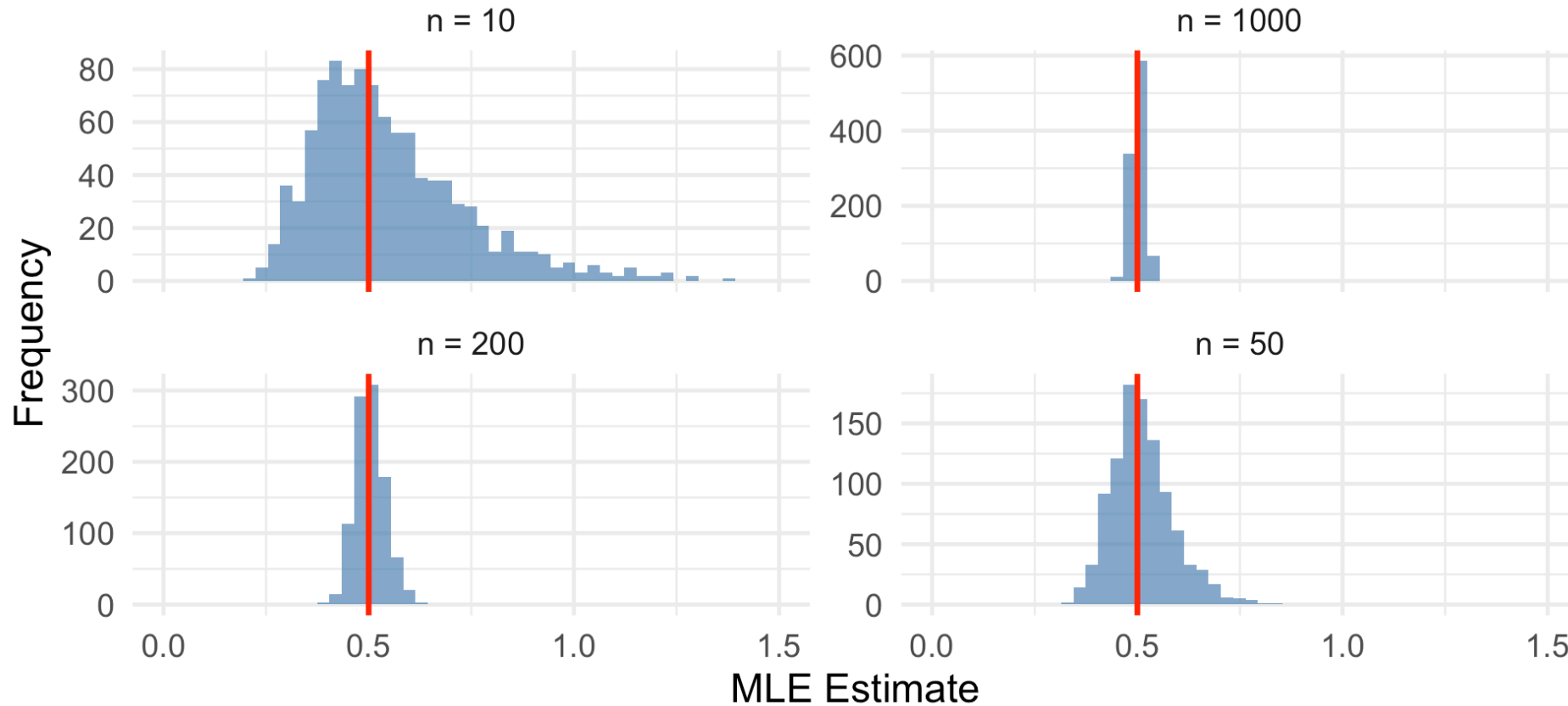
- Variance $\rightarrow 0$ (estimates get more precise)
- Bias $\rightarrow 0$ (estimates get more accurate)



Visualizing Consistency

Consistency: MLE Concentrates Around True Value

Exponential MLE: $\hat{\lambda} = 1/\bar{X}$, True $\lambda = 0.5$



As n increases, the distribution “shrinks” around the truth!

Part 3: Finding the Best Estimator

The MVUE: Minimum Variance Unbiased Estimator

! Definition (Devore 7.1)

The **MVUE** of θ is the estimator that:

1. Is unbiased: $E(\hat{\theta}) = \theta$
2. Has smaller variance than **any** other unbiased estimator

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Why not just minimize MSE?

- MSE depends on the true θ (which we don't know!)
- A biased estimator might be better at some θ values, worse at others
- MVUE provides a clean, universal optimality criterion

The Cramér-Rao Lower Bound (CRLB)

! Theorem: CRLB (Devore 7.4)

For any unbiased estimator T of θ :

$$\text{Var}(T) \geq \frac{1}{nI(\theta)}$$

where $I(\theta)$ is the **Fisher Information**:

$$I(\theta) = E \left[-\frac{\partial^2}{\partial \theta^2} \ln f(X; \theta) \right]$$

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Interpretation:

- The CRLB is a **floor** — no unbiased estimator can do better
- If an estimator achieves the CRLB, it's the **MVUE**
- Fisher Information measures the “curvature” of the log-likelihood

Fisher Information: The Intuition

Key Insight: Fisher Information measures how much the log-likelihood changes as θ changes.

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High Fisher Information:

- Sharp, peaked likelihood $\rightarrow$ easy to pinpoint θ
- Small variance possible

Low Fisher Information:

- Flat likelihood $\rightarrow$ hard to distinguish θ values
- Large variance unavoidable

Efficiency: How Close to Optimal?

! Definition: Efficiency

An unbiased estimator T is **efficient** if:

$$\text{Var}(T) = \frac{1}{nI(\theta)} = \text{CRLB}$$

More generally, the **efficiency** of T is:

$$\text{Efficiency}(T) = \frac{\text{CRLB}}{\text{Var}(T)} \leq 1$$

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Example: Sample median for normal data

- Variance $\approx 1.57 \cdot \sigma^2/n$
- CRLB $= \sigma^2/n$
- Efficiency $= 1/1.57 \approx 0.64$ (64% efficient)

Key Results: Efficient Estimators

Distribution	Parameter	MVUE	CRLB	Efficient?
Bernoulli(p)	p	$\bar{X}$	$\frac{p(1-p)}{n}$	✓
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i Note on Exponential(λ): Why No Efficient Unbiased Estimator Exists

For the exponential distribution with **rate parameter** λ :

- **MLE:** $\hat{\lambda}_{MLE} = 1/\bar{X}$ is **biased** – $E(1/\bar{X}) = \frac{n\lambda}{n-1} \neq \lambda$
- **MVUE:** The bias-corrected estimator $\hat{\lambda}_{MVUE} = \frac{n-1}{\sum X_i} = \frac{n-1}{n} \cdot \frac{1}{\bar{X}}$
- **Variance of MVUE:** $\text{Var}(\hat{\lambda}_{MVUE}) = \frac{\lambda^2}{n-2}$ (for $n > 2$)
- **CRLB:** $\frac{\lambda^2}{n}$

Since $\frac{\lambda^2}{n-2} > \frac{\lambda^2}{n}$, the MVUE does **not** achieve the CRLB. No unbiased estimator can!

Alternatively: For the **mean parameter** $\mu = 1/\lambda$, the sample mean $\bar{X}$ is the efficient MVUE.

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i Note on Uniform[0, θ]

Part 4: Sufficient Statistics — The Key to Everything

Why Sufficiency Matters: The Central Concept

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Answer: Sufficient statistics answer a fundamental question:

What is the minimum amount of information we need to keep from our data to estimate θ ?

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Answer: Sufficient statistics answer a fundamental question:

What is the minimum amount of information we need to keep from our data to estimate θ ?

If a statistic T is sufficient, then:

- We can throw away the raw data and just keep T
- We lose **no information** about θ
- All optimal estimators will be functions of T

Intuition: What Does “Sufficient” Really Mean?

Scenario: You observe n coin flips and want to estimate p (probability of heads).

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Summary: 6 heads out of 10

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Summary: 6 heads out of 10

Question: Does knowing the exact sequence (HTHHTHTTTHH) tell you more about p than just knowing “6 out of 10”?

Answer: No! The total number of heads captures all the information about p .

This is why $T = \sum X_i$ is **sufficient** for p in the Bernoulli model.

Formal Definition of Sufficiency

! Definition (Devore 7.3)

A statistic $T = t(X_1, \dots, X_n)$ is **sufficient** for θ if:

$$P(X_1 = x_1, \dots, X_n = x_n \mid T = t) \text{ does not depend on } \theta$$

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In words: Given the value of T , the conditional distribution of the data doesn't involve θ .

Intuition: Once you know T , learning the raw data tells you nothing new about θ .

The Neyman Factorization Theorem

Computing conditional distributions is hard. The **factorization theorem** gives us a shortcut!

! Theorem (Neyman-Fisher Factorization)

T is sufficient for θ if and only if the joint pdf/pmf can be written as:

$$f(x_1, \dots, x_n; \theta) = g(T; \theta) \cdot h(x_1, \dots, x_n)$$

where:

- g involves θ and depends on data **only through** T
- h depends on data but **not on** θ

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- h depends on data but **not on** θ

Strategy: Look for parts involving θ and group them with a function of the data.

Example: Finding the Sufficient Statistic for Poisson

Setup: $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poisson}(\mu)$

Joint pmf:

$$f(x_1, \dots, x_n; \mu) = \prod_{i=1}^n \frac{e^{-\mu} \mu^{x_i}}{x_i!} = \frac{e^{-n\mu} \mu^{\sum x_i}}{\prod x_i!}$$

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Factor it:

$$= \underbrace{e^{-n\mu} \mu^T}_{g(T; \mu)} \cdot \underbrace{\frac{1}{\prod x_i!}}_{h(x_1, \dots, x_n)}$$

where $T = \sum x_i$.

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 Result

$T = \sum_{i=1}^n X_i$ is sufficient for μ .



Example: Uniform[0, θ] – A Surprising Result

Setup: $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Uniform}[0, \theta]$

Joint pdf:

$$f(x_1, \dots, x_n; \theta) = \frac{1}{\theta^n} \cdot I(0 \leq \min(x_i)) \cdot I(\max(x_i) \leq \theta)$$

Example: Uniform[0, θ] – A Surprising Result

Setup: $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Uniform}[0, \theta]$

Joint pdf:

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Factor it:

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! Key Result

$T = \max(X_i)$ is sufficient for θ .

Notice: $\bar{X}$ is **NOT** sufficient for the uniform distribution!

Why Isn't $\bar{X}$ Sufficient for Uniform?

```
1 # Two datasets with same mean but different information about  $\theta$ 
2 data_A <- c(2, 4, 6, 8) # mean = 5, max = 8
3 data_B <- c(1, 4, 5, 10) # mean = 5, max = 10
4
5 cat("Dataset A: mean =", mean(data_A), ", max =", max(data_A), "\n")
```

Dataset A: mean = 5 , max = 8

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1 cat("Dataset B: mean =", mean(data_B), ", max =", max(data_B), "\n")
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Dataset B: mean = 5 , max = 10

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```

Dataset B: mean = 5 , max = 10

Both have $\bar{X} = 5$, but:

- Dataset A tells us $\theta \geq 8$
- Dataset B tells us $\theta \geq 10$

Knowing only $\bar{X}$ loses critical information about θ !

Sufficient Statistics: Summary Table

Distribution	Parameter(s)	Sufficient Statistic(s)
Bernoulli(p)	p	$\sum X_i$
Poisson(μ)	μ	$\sum X_i$
Exponential(λ)	λ	$\sum X_i$
Normal(μ , known σ^2)	μ	$\bar{X}$
Normal(μ , σ^2)	(μ, σ^2)	$(\bar{X}, S^2)$ or $(\sum X_i, \sum X_i^2)$
Uniform[0, θ]	θ	$\max(X_i)$

Why Sufficiency Matters: The Big Picture

! Three Key Connections

1. MLEs are functions of sufficient statistics

- You don't lose anything by working with T instead of raw data

2. MVUEs are functions of sufficient statistics (Rao-Blackwell)

- The best unbiased estimator depends only on T

3. If T is complete and sufficient, any unbiased function of T is the MVUE (Lehmann-Scheffé)

- Finding the MVUE becomes much easier!

The Rao-Blackwell Theorem: Improving Estimators

! Theorem (Rao-Blackwell)

If U is any unbiased estimator of $\tau(\theta)$ and T is sufficient for θ , then:

$$U^* = E(U \mid T)$$

is also unbiased, and $\text{Var}(U^*) \leq \text{Var}(U)$.

The Rao-Blackwell Theorem: Improving Estimators

! Theorem (Rao-Blackwell)

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In plain English: Conditioning on a sufficient statistic **never hurts** and usually **helps**.

Practical implication: If your estimator doesn't use all the information in T , you can always improve it!

Medical Example: Rao-Blackwell in Action

Problem: Estimate $e^{-\mu} = P(\text{no defects})$ for $\text{Poisson}(\mu)$ data.

Crude estimator: $U = I(X_1 = 0)$ (is the first observation zero?)

► Code

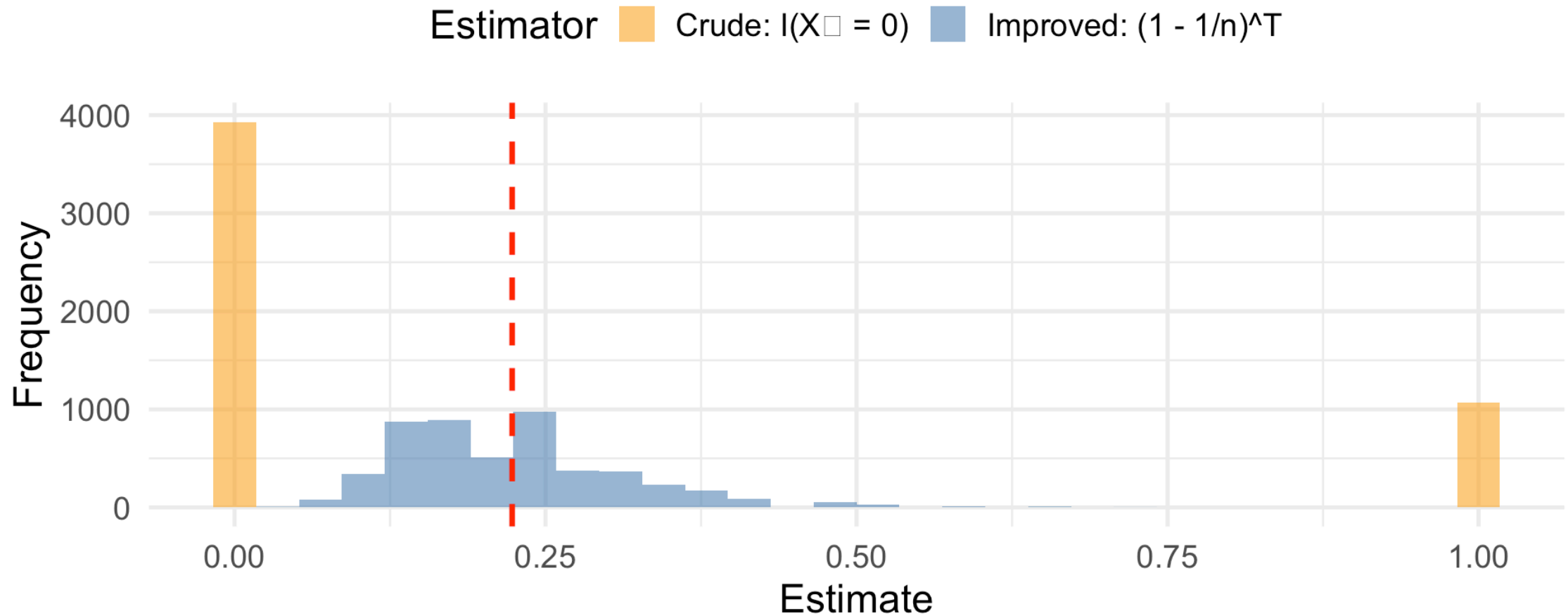
```
# A tibble: 5 × 2
  name          value
  <chr>        <dbl>
1 True e(-μ)    0.223
2 Crude: E[U]   0.214
3 Crude: Var    0.168
4 Improved: E[U*] 0.223
5 Improved: Var  0.0082
```

The Rao-Blackwell estimator has **much lower variance!**

Visualization: Rao-Blackwell Improvement

Rao-Blackwell: Variance Reduction Through Sufficiency

True $e^{(-\mu)} = 0.223$



Completeness: The Final Piece

! Definition: Complete Statistic

T is **complete** if for any function g :

$$E_{\theta}[g(T)] = 0 \text{ for all } \theta \implies P_{\theta}(g(T) = 0) = 1 \text{ for all } \theta$$

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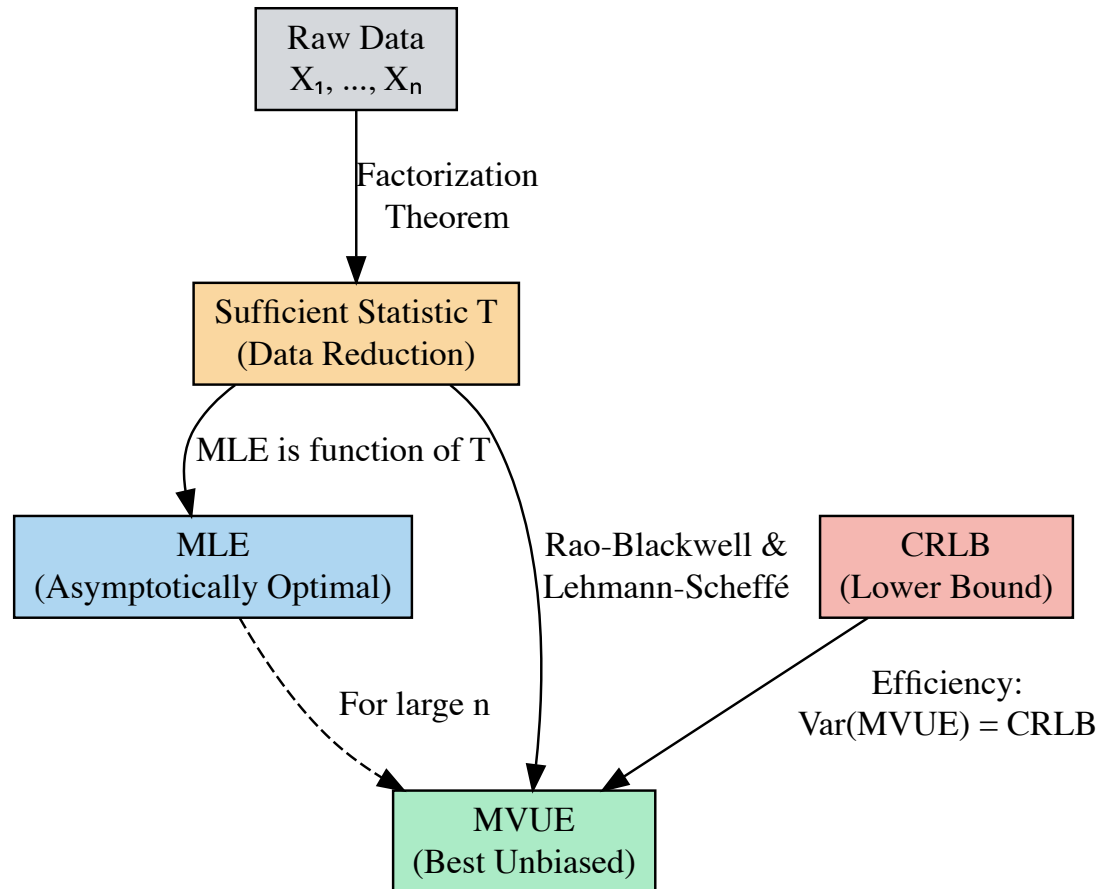
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💡 Lehmann-Scheffé Theorem

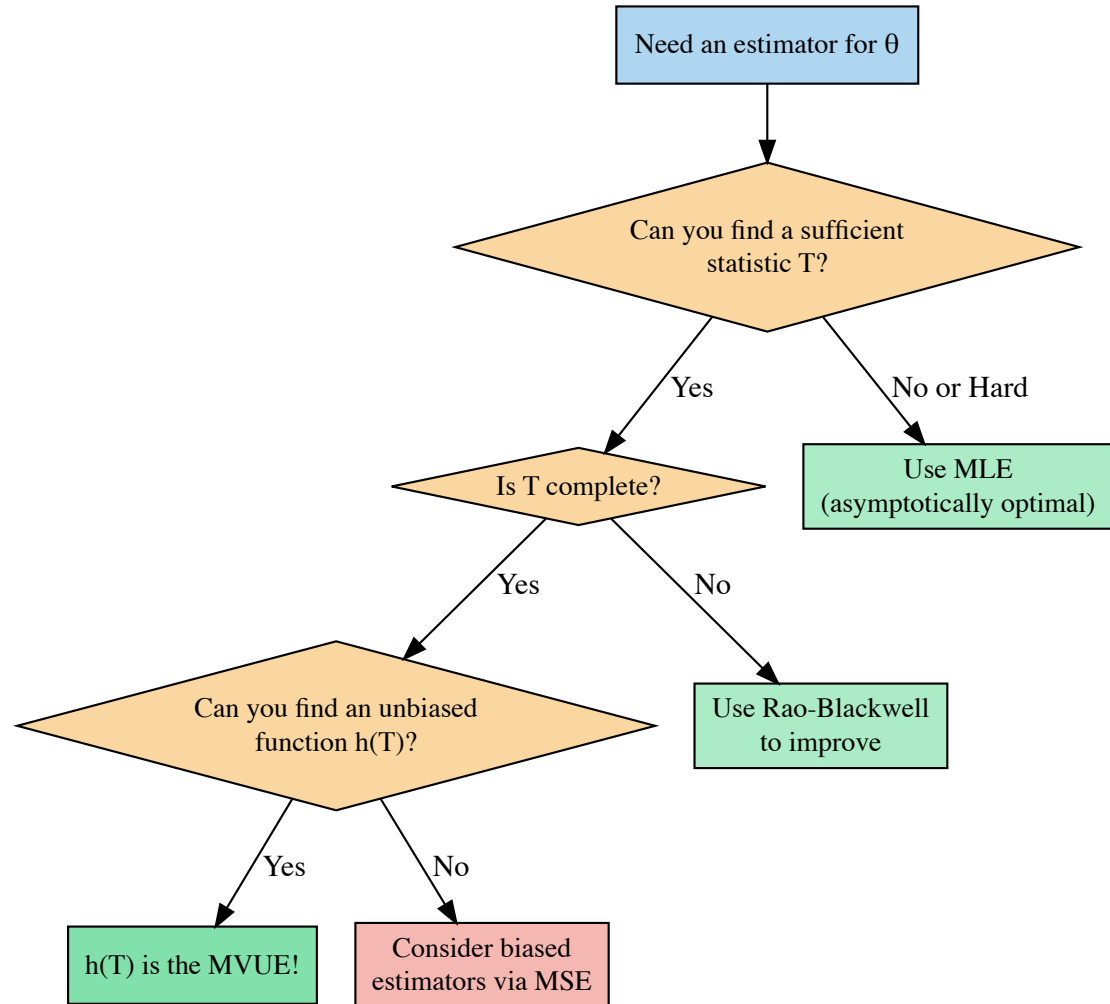
If T is **complete** and **sufficient**, and $h(T)$ is unbiased for $\tau(\theta)$, then $h(T)$ is the **unique MVUE** of $\tau(\theta)$.

Part 5: Putting It All Together

The Complete Picture: How Everything Connects



Decision Tree: Finding the Best Estimator



(Note: Claude AI put these flow charts together for me, thanks Claude!)

Summary: Key Formulas to Remember

Concept	Formula
Bias	$E(\hat{\theta}) - \theta$
MSE	$\text{Var}(\hat{\theta}) + \text{Bias}^2$
Standard Error	$\sqrt{\text{Var}(\hat{\theta})}$
Fisher Information	$I(\theta) = E[-\ell''(\theta)]$
CRLB	$\frac{1}{nI(\theta)}$
Efficiency	$\frac{\text{CRLB}}{\text{Var}(\hat{\theta})}$
Consistency	$\text{MSE} \rightarrow 0 \text{ as } n \rightarrow \infty$

Summary: Key Results by Distribution

Distribution	MLE	MME	Sufficient Stat	MVUE of Mean
Bernoulli(p)	$\bar{X}$	$\bar{X}$	$\sum X_i$	$\bar{X}$
Poisson(μ)	$\bar{X}$	$\bar{X}$	$\sum X_i$	$\bar{X}$
Exponential(λ)	$1/\bar{X}$	$1/\bar{X}$	$\sum X_i$	$\bar{X}$ (for mean)
Normal(μ, σ^2)	$\bar{X},$ $\frac{1}{n} \sum (X_i - \bar{X})^2$	Same	$(\bar{X}, S^2)$	$\bar{X}$
Uniform[0, θ]	$\max(X_i)$	$2\bar{X}$	$\max(X_i)$	$\frac{n+1}{n} \max(X_i)$

Common Misconceptions to Avoid

 Pitfall 1: “Unbiased is always better”

Reality: A biased estimator with low variance can have lower MSE!

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⚠ Pitfall 1: “Unbiased is always better”

Reality: A biased estimator with low variance can have lower MSE!

⚠ Pitfall 2: “MLE = Sufficient Statistic”

Reality: MLE is a **function of** sufficient statistics, not the sufficient statistic itself.

⚠ Pitfall 3: “ $\bar{X}$ is always sufficient”

Reality: $\bar{X}$ is sufficient for location parameters (normal, exponential) but NOT for scale parameters like in Uniform $[0, \theta]$.

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Chapters 6-7.
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Chapter 6.

Questions?

Thank you!